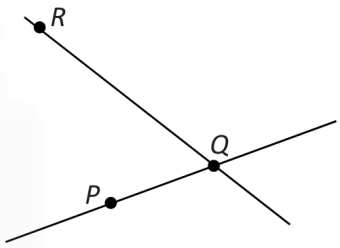
**Question 2** **(25 marks)**

The diagram shows the line *PQ* and the line *QR*.
The co-ordinates of the points are *P*(4, 2), *Q*(8, 5) and *R*(2, 11).




**(a)** Find the slope of *PQ*.

**(b)** Find the equation of the line *PQ*.
Give your answer in the form $ax + by + c = 0$, where $a, b, c \in \mathbb{Z}$.

**(c)** Write down the slope of any line perpendicular to *PQ*.

Slope =

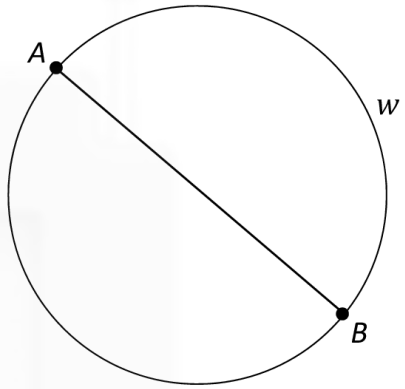
**(d)** Find the area of the triangle *PQR*.

| | | |
|---|---|---|
| **(a)** | $m = \frac{y_2 - y_1}{x_2 - x_1}$<br>$= \frac{5-2}{8-4}$<br>$= \frac{3}{4}$ | **Scale 10C(0, 4, 6, 10)**<br>*Low Partial Credit:*<br>• Work of merit<br>• Formula with some correct substitution<br><br>*High Partial Credit:*<br>• Formula substituted correctly<br><br>*Full Credit:*<br>• Correct answer without work<br><br>*Misreading:*<br>• Finds slope of *PR* or *QR* |
| **(b)** | $y - y_1 = m(x - x_1)$<br>$y - 2 = \frac{3}{4}(x - 4)$<br>$3x - 4y - 4 = 0$ | **Scale 5C (0, 2, 3, 5)**<br>*Low Partial Credit:*<br>• Formula with some correct or consistent substitution<br><br>*High Partial Credit:*<br>• Formula substituted correctly or consistently<br>• One incorrect substitution followed by correct solution<br>• Answer not in the required format<br><br>**Note:**<br>Accept<br>$-3x + 4y + 4 = 0$ |
| **(c)** | $-\frac{4}{3}$ | **Scale 5B (0, 2, 5)**<br>*Mid Partial Credit:*<br>• Work of merit<br><br>*Full Credit:*<br>• Correct answer without work |

| | | |
|---|---|---|
| **(d)** | $(4,2)\ (8,5)\ (2,11)$<br>$\downarrow \quad \downarrow \quad \downarrow$<br>$(0,0)\ (4,3)\ (-2,9)$<br>$= \frac{1}{2}\lvert x_1y_2 - x_2y_1 \rvert$<br>$= \frac{1}{2}\lvert (4)(9) - (-2)(3) \rvert$<br>$= \frac{1}{2}\lvert 36 + 6 \rvert$<br>$= 21$ square units | **Scale 5D (0, 2, 3, 4, 5)**<br>*Low Partial Credit:*<br>• Translation of any relevant point to $(0,0)$<br>• Area of triangle formula with some substitution<br><br>*Mid Partial Credit:*<br>• Area of triangle formula substituted fully with 2 from P,Q, and R<br><br>*High Partial Credit:*<br>• Formula substituted correctly with translated points<br>• Area of triangle found from formula substituted fully with 2 from *P,Q, R*<br>• An incorrect translation applied correctly |

**Question 4** **(25 marks)**

The points $A(1, 8)$ and $B(9, 0)$ are the end-points of a diameter of the circle $w$, as shown in the diagram.




**(a)** Find the co-ordinates of the centre of $w$.

**(b)** Find the length of the radius of $w$. Give your answer in the form $p\sqrt{q}$, where $p, q \in \mathbb{N}$.

**(c)** Hence write down the equation of the circle $w$.

**(d)** Find the equation of the line that is a tangent to the circle $w$ at $A$.
Give your answer in the form $ax + by + c = 0$, where $a, b$, and $c \in \mathbb{Z}$.

| Q4 | Model Solution – 25 Marks | Marking Notes |
|---|---|---|
| **(a)** | $\text{Centre} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$<br>$= \left(\frac{1+9}{2}, \quad \frac{8+0}{2}\right)$<br>$= (5, 4)$<br><br>**<u>Method 2 : Translation</u>**<br><br>(1,8) →(Centre)<br><br>X ↑4, y ↓4<br><br>(1,8) →(5,4) | **Scale 5B (0, 2, 5)**<br>*Partial Credit:*<br>• Mid-point formula with some correct substitution.<br>• Leaves answer in fraction form.<br>• Indicates centre.<br>• Correct translation identified.<br><br>*Full Credit:*<br>• Correct answer without work. |
| **(b)** | $\lvert BC \rvert = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$<br>$= \sqrt{(9-5)^2 + (0-4)^2}$<br>$= \sqrt{(4)^2 + (-4)^2}$<br>$= \sqrt{16 + 16}$<br>$= \sqrt{32}$<br>$= 4\sqrt{2}$ units | **Scale 5C(0, 2, 3, 5)**<br>*Low Partial Credit:*<br>• Correct formula with some correct or consistent substitution.<br><br>*High Partial Credit:*<br>• Formula fully substituted.<br>• Correct answer without work. |
| **(c)** | $(x-h)^2 + (y-k)^2 = r^2$<br>$(x-5)^2 + (y-4)^2 = \sqrt{32}^2$<br>$(x-5)^2 + (y-4)^2 = 32$<br>**Or**<br>$x^2 + y^2 - 10x - 8y + 9 = 0$ | **Scale 10C(0, 3, 7, 10)**<br>*Low Partial Credit:*<br>• Correct formula with some correct or consistent substitution.<br><br>*High Partial Credit:*<br>• Answer as<br>$(x-5)^2 + (y-4)^2 = \sqrt{32}^2$<br><br>*Full Credit:*<br>• Correct or consistent answer without work. |

| | | |
|---|---|---|
| **(d)** | $m_r = -1$<br>$m_T = 1$<br>$y - 8 = 1(x - 1)$<br>$x - y + 7 = 0$ | **Scale 5C (0, 2, 3, 5)**<br>*Low Partial Credit:*<br>• Correct relevant formula with some correct substitution.<br>• Finds slope of radius and stops.<br><br>*High Partial Credit:*<br>• Eq. of line formula fully substituted.<br><br>*Full Credit:*<br>• Answer as $(-x + y - 7 = 0)$.<br><br>*Misreading:*<br>• Equation of tangent through B $(x - y - 9 = 0)$. |

**Question 8** **(40 marks)**

The diagram below shows the triangles *CBA* and *CDE*.

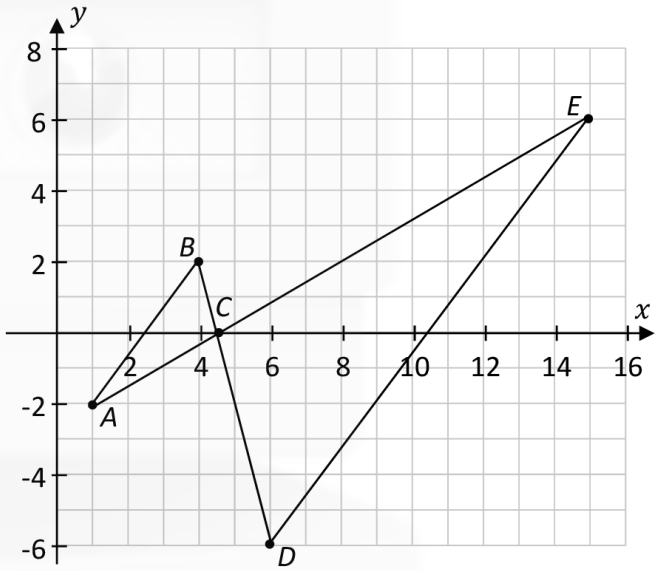
**(a)** The coordinates of *C* are $(4{\cdot}5, 0)$.
From the diagram, write down the coordinates of the points *A*, *B*, *D*, and *E*.

*A* = ( , )

*B* = ( , )

*D* = ( , )

*E* = ( , )




**(b)** Show, using slopes, that the line segments [*AB*] and [*DE*] are parallel.

**(c)** **(i)** Show that the area of the triangle *CBA* is 4 square units.

**(ii)** Find $|AB|$, the distance from *A* to *B*.

**(iii)** The triangle *CDE* is an enlargement of the triangle *CBA*.
Given that $|DE| = 15$ units, find the scale factor of the enlargement.

Scale factor =

**(iv)** Use this scale factor to find the area of the triangle *CDE*.

| | | |
|---|---|---|
| **(a)** | $A = (1, -2)$ $\quad$ $B = (4, 2)$<br>$D = (6, -6)$ $\quad$ $E = (15, 6)$ | **Scale 10C (0, 3, 7, 10)**<br>*Low Partial Credit:*<br>• ONE correct point.<br>*High Partial Credit:*<br>• THREE correct points.<br>• All answers written consistently as $(y,\ x)$ |
| **(b)** | $m_{AB} = \frac{4}{3}$<br>$m_{DE} = \frac{12}{9}$ **or** $\frac{4}{3}$<br>**Conclusion:**<br>Parallel, or slopes are the same. | **Scale 10D (0, 4, 5, 7, 10)**<br>*Low Partial Credit:*<br>• Any mention of same slopes or similar.<br>• Slope formula written with some relevant substitution.<br>• Rise over Run (Written or indicated).<br>*Mid Partial Credit:*<br>• Slope formula fully substituted for one slope.<br>• One correct slope.<br>*High Partial Credit:*<br>• Both slopes but no conclusion.<br>• One error in substitution followed by correct calculation and consistent conclusion.<br>*Full credit:*<br>• Equal slopes and conclusion. |
| **(c)**<br>**(i)** | $A = (1, -2)$ $B = (4, 2)$ $C = (4{\cdot}5, 0)$<br>$\downarrow$ $\downarrow$ $\downarrow$<br>$A = (0, 0)$ $B = (3, 4)$ $C = (3{\cdot}5, 2)$<br>$\text{Area} = \frac{1}{2}[(3)(2) - (3{\cdot}5)(4)]$<br>$\text{Area} = \frac{1}{2}\lvert(6) - (14)\rvert$<br>$= \frac{1}{2}\lvert(-8)\rvert$<br>$= 4$ | **Scale 5C (0, 2, 3, 5)**<br>*Low Partial Credit:*<br>• Correct relevant formula with some correct/consistent substitution.<br>• One point translated to (0, 0).<br>*High Partial Credit:*<br>• Area= $\frac{1}{2}\lvert x_1y_2 - x_2y_1\rvert$ fully substituted with translated points.<br>• Error in translation and area calculated correctly.<br>• No translation but uses TWO of the points A, B, C and area calculated correctly.<br>• One error in substitution followed by correct calculation. |

| | | |
|---|---|---|
| **(c) (ii)** | $\lvert AB\rvert = \sqrt{3^2 + 4^2} = 5$ units | **Scale 5C (0, 2, 3, 5)**<br>*Low Partial Credit:*<br>• Correct relevant formula with some substitution.<br>• Any work of merit indicated in this part.<br><br>*High Partial Credit:*<br>• Correct substitution without calculation.<br>• One error in substitution followed by correct calculation.<br>• Correct answer without work. |
| **(c) (iii)** | $\frac{15}{5} = 3 = k$ | **Scale 5B (0, 3, 5)**<br>*Partial Credit:*<br>• Use of answer to part (c)(ii).<br>• Scale factor formula written or indicated.<br>• Attempts to calculate $\lvert AB\rvert$ in this part.<br><br>*Full Credit:*<br>• Correct answer without work. |
| **(c) (iv)** | Area $= 4 \times 3^2 = 36$ units$^2$ | **Scale 5C (0, 2, 3, 5)**<br>*Low Partial Credit:*<br>• Work of merit with answer from (c)(iii).<br>• Work of merit with answer from (c)(i).<br><br>*High Partial Credit:*<br>• Area as $3 \times 4 = 12$ or equivalent.<br><br>*Full Credit:*<br>• Correct answer without work. |